Data Structures and Algorithms CA1 Proposal

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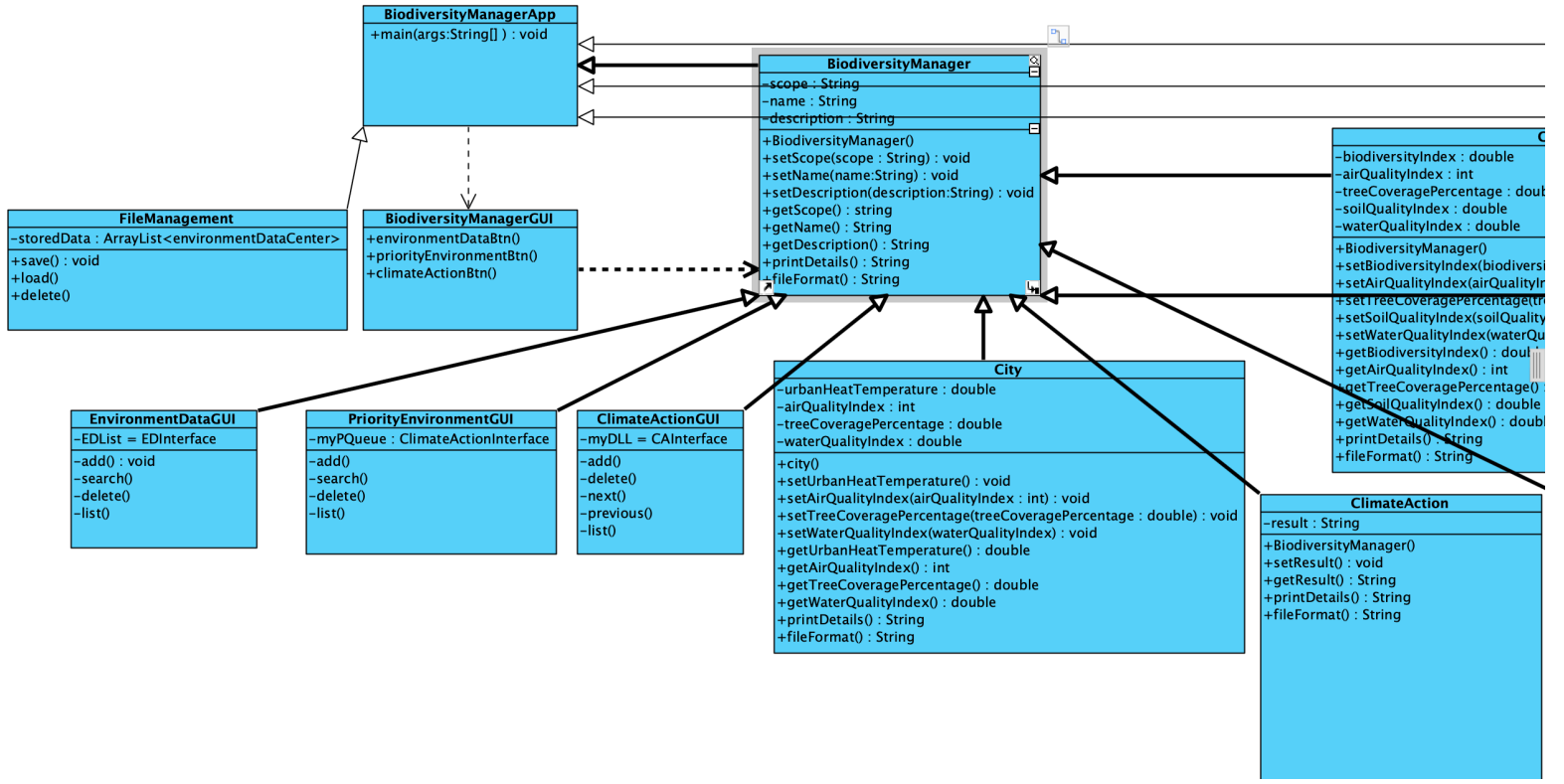
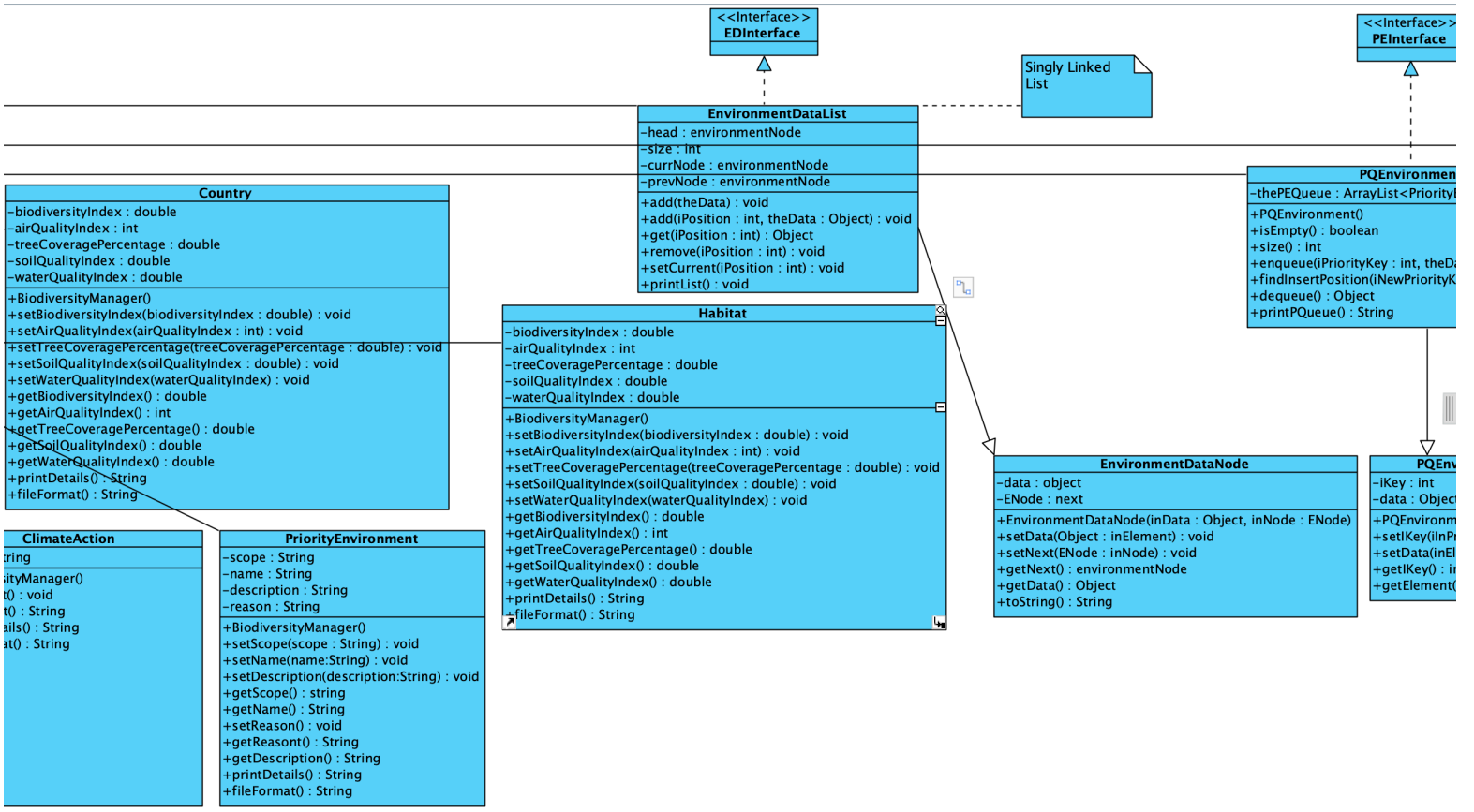
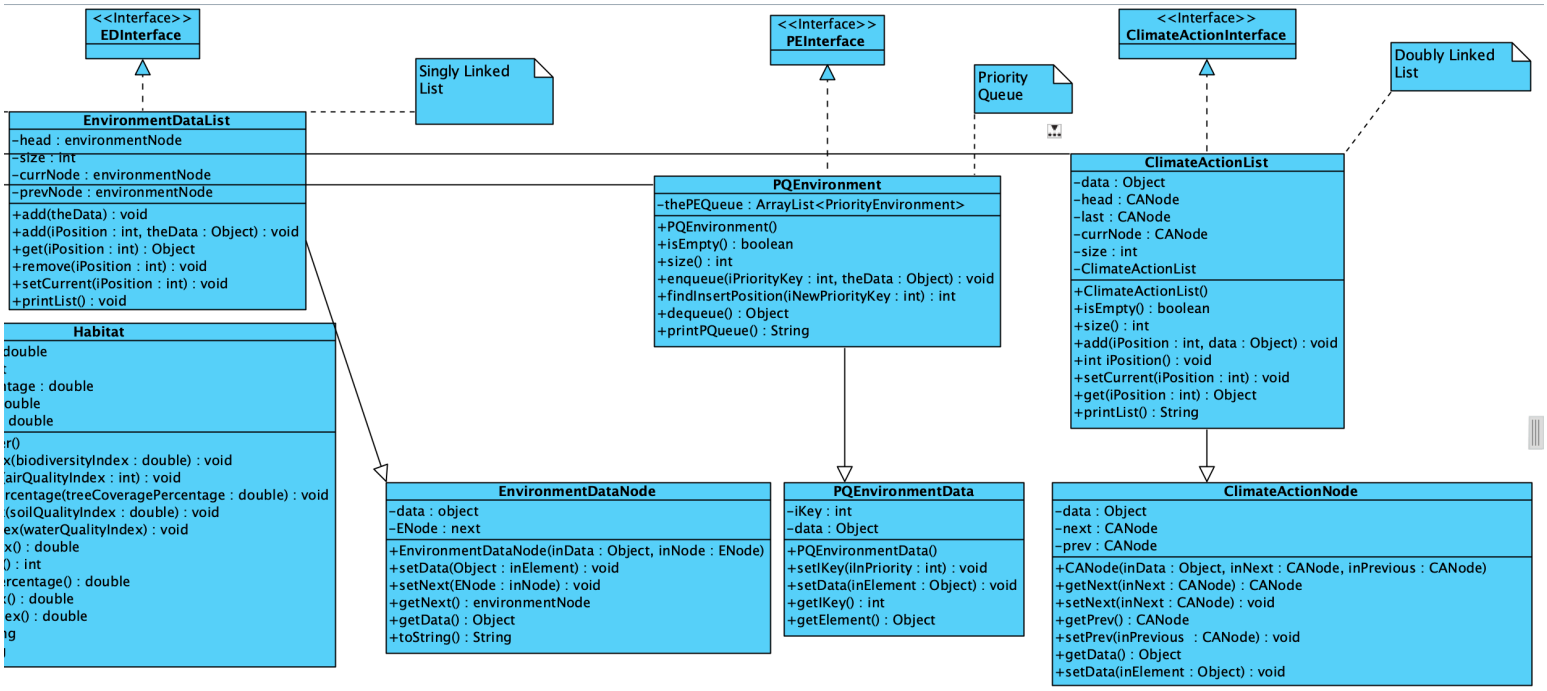
The theme I have chosen from smart docklands pilot call themes for my application to tackle and propose a range of varied solutions to is “Urban Greening/Biodiversity”.
My application BiodiversityManager will allow the user to keep, manage data and draw conclusions from the data collected from sensors, internet of things and other data driven monitor tools.

The BiodiversityManager applications main page will have three sections one will allow the user to use “Environment Data” section to choose the scope which is whether it is a city e.g Dublin, a country e.g Ireland or specific habitat they are focusing on at the moment. Along with that they can enter the specific name of their scope such as “Dublin”, “Ireland”, “Atlantic woodlands” respectively. The user will also enter a description of the general state of the environment. Then they can enter the biodiversity index, the urban heat temperature, the air quality index, the tree coverage percentage, the soil quality index and water quality index can all be saved. For countries the biodiversity index, the air quality index, the tree coverage percentage, the soil quality index and the water quality index can be added. For a city the urban heat temperature, the air quality index and the tree coverage percentage. For a habitat the biodiversity index, the air quality index, the tree coverage percentage, the soil quality index and the water quality index. The BiodiversityManager also allows the ability to search through the added data and delete data by using the scope and name.
The BiodiversityManager applications will also allow the user through "Priority Environment” section to prioritise certain habitats for example, add a description of why it is of concern rather than having them in the order they were added.

The BiodiversityManager “ClimateAction” will also allow the user to use enter a specific habitat for example, enter solutions that have been applied to it and the result that the action. This will also include the ability to view the data forward and backward.
The environmental data be can be stored on BiodiversityManager analysed and allow solutions to be developed in order tackle climate action goals and sustain biodiversity in one place. BiodiversityManager can store various climate action goals, such as enhancing the sustainability habitats or the sustainability of the overall environment of an area such as city, and help in managing the biodiversity of a country to even a small village. The application helps in the ultimate goal is of testing the impact that climate actions have and the local resilience of these environment by storing and allowing easy viewing of the information. The use of the insights gained from the data stored and analysed on BiodiversityManager can help in urban greening effort and biodiversity preservation. The integration of data driven tool by BiodiversityManager allows large amount of data collected from internet of things and sensors to be consistent, and available. Thus enhancing the ability to plan and develop proper solutions for resolving challenges to urban greening and biodiversity.


BiodiversityManagerApp
+main(args:String[]) : void
BiodiversityManager
-scope : String
-name : String
-description : String
+BiodiversityManager()
+setScope(scope : String) : void
+setName(name:String) : void
+setDescription(description:String) : void
+getScope() : string
+getName() : String
+getDescription() : String
+printDetails() : String
+fileFormat() : String
FileManagement
-storedData : ArrayList<environmentDataCenter>
+save() : void
+load()
+delete()
BiodiversityManagerGUI
+environmentDataBtn()
+priorityEnvironmentBtn()
+climateActionBtn()
-biodiversityIndex : double
-airQualityIndex : int
-soilQualityIndex : double
-waterQualityIndex : double
+BiodiversityManager()
+fileFormat() : String
EnvironmentDataGUI
-EDList = EDInterface
-add() : void
-search()
-delete()
-list()
PriorityEnvironmentGUI
-myPQueue : ClimateActionInterface
-add()
-search()
-delete()
-list()
ClimateActionGUI
-myDLL = CAInterface
-add()
-delete()
-next()
-previous()
-list()
City
-urbanHeatTemperature : double
-airQualityIndex : int
-treeCoveragePercentage : double
-waterQualityIndex : double
+city()
+setUrbanHeatTemperature() : void
+setAirQualityIndex(airQualityIndex : int) : void
+setTreeCoveragePercentage(treeCoveragePercentage : double) : void
+setWaterQualityIndex(waterQualityIndex) : void
+getUrbanHeatTemperature() : double
+getAirQualityIndex() : int
+getTreeCoveragePercentage() : double
+getWaterQualityIndex() : double
+printDetails() : String
+fileFormat() : String
ClimateAction
-result : String
+BiodiversityManager()
+setResult() : void
+getResult() : String
+printDetails() : String
+fileFormat() : String


<<Interface>>
EDInterface
<<Interface>>
PEInterface
Singly Linked List
EnvironmentDataList
-head : environmentNode
-size : int
-currNode : environmentNode
-prevNode : environmentNode
+add(theData) : void
+add(iPosition : int, theData : Object) : void
+get(iPosition : int) : Object
+remove(iPosition : int) : void
+setCurrent(iPosition : int) : void
+printList() : void
Country
-biodiversityIndex : double
-airQualityIndex : int
-treeCoveragePercentage : double
-soilQualityIndex : double
-waterQualityIndex : double
+BiodiversityManager()
+setBiodiversityIndex(biodiversityIndex : double) : void
+setAirQualityIndex(airQualityIndex : int) : void
+setTreeCoveragePercentage(treeCoveragePercentage : double) : void
+setSoilQualityIndex(soilQualityIndex : double) : void
+setWaterQualityIndex(waterQualityIndex) : void
+getBiodiversityIndex() : double
+getAirQualityIndex() : int
+getTreeCoveragePercentage() : double
+getSoilQualityIndex() : double
+getWaterQualityIndex() : double
+printDetails() : String
+fileFormat() : String
Habitat
-biodiversityIndex : double
-airQualityIndex : int
-treeCoveragePercentage : double
-soilQualityIndex : double
-waterQualityIndex : double
+BiodiversityManager()
+setBiodiversityIndex(biodiversityIndex : double) : void
+setAirQualityIndex(airQualityIndex : int) : void
+setTreeCoveragePercentage(treeCoveragePercentage : double) : void
+setSoilQualityIndex(soilQualityIndex : double) : void
+setWaterQualityIndex(waterQualityIndex) : void
+getBiodiversityIndex() : double
+getAirQualityIndex() : int
+getTreeCoveragePercentage() : double
+getSoilQualityIndex() : double
+getWaterQualityIndex() : double
+printDetails() : String
+fileFormat() : String
EnvironmentDataNode
-data : object
-ENode : next
+EnvironmentDataNode(inData : Object, inNode : ENode)
+setData(Object : inElement) : void
+setNext(ENode : inNode) : void
+getNext() : environmentNode
+getData() : Object
+toString() : String
-thePEQueue : ArrayList<Priority
+PQEnvironment()
+isEmpty() : boolean
+size() : int
+enqueue(iPriorityKey : int, theD
+findInsertPosition(iNewPriorityK
+dequeue() : Object
+printPQueue() : String
ClimateAction
PriorityEnvironment
-scope : String
-name : String
-description : String
-reason : String
+BiodiversityManager()
+setScope(scope : String) : void
+setName(name:String) : void
+setDescription(description:String) : void
+getScope() : string
+getName() : String
+setReason() : void
+getReasont() : String
+getDescription() : String
+printDetails() : String
+fileFormat() : String


<<Interface>>
EDInterface
<<Interface>>
PEInterface
<<Interface>>
ClimateActionInterface
Singly Linked List
Priority Queue
Doubly Linked List
EnvironmentDataList
-head : environmentNode
-size : int
-currNode : environmentNode
-prevNode : environmentNode
+add(theData) : void
+add(iPosition : int, theData : Object) : void
+get(iPosition : int) : Object
+remove(iPosition : int) : void
+setCurrent(iPosition : int) : void
+printList() : void
PQEnvironment
-thePEQueue : ArrayList<PriorityEnvironment>
+PQEnvironment()
+isEmpty() : boolean
+size() : int
+enqueue(iPriorityKey : int, theData : Object) : void
+findInsertPosition(iNewPriorityKey : int) : int
+dequeue() : Object
+printPQueue() : String
ClimateActionList
-data : Object
-head : CANode
-last : CANode
-currNode : CANode
-size : int
-ClimateActionList
+ClimateActionList()
+isEmpty() : boolean
+size() : int
+add(iPosition : int, data : Object) : void
+int iPosition() : void
+setCurrent(iPosition : int) : void
+get(iPosition : int) : Object
+printList() : String
Habitat
double
tage : double
ouble
double
r()
x(biodiversityIndex : double) : void
airQualityIndex : int) : void
rcentage(treeCoveragePercentage : double) : void
(soilQualityIndex : double) : void
ex(waterQualityIndex) : void
x() : double
) : int
rcentage() : double
() : double
ex() : double
g
EnvironmentDataNode
-data : object
-ENode : next
+EnvironmentDataNode(inData : Object, inNode : ENode)
+setData(Object : inElement) : void
+setNext(ENode : inNode) : void
+getNext() : environmentNode
+getData() : Object
+toString() : String
PQEnvironmentData
-iKey : int
-data : Object
+PQEnvironmentData()
+setIKey(iInPriority : int) : void
+setData(inElement : Object) : void
+getIKey() : int
+getElement() : Object
ClimateActionNode
-data : Object
-next : CANode
-prev : CANode
+CANode(inData : Object, inNext : CANode, inPrevious : CANode)
+getNext(inNext : CANode) : CANode
+setNext(inNext : CANode) : void
+getPrev() : CANode
+setPrev(inPrevious : CANode) : void
+getData() : Object
+setData(inElement : Object) : void

References

Smartdoc (2026). *Smart Docklands Call for Pilots 2025: Deadline Extended - smartdocklands*. [online] smartdocklands. Available at: https://smartdocklands.ie/smart-docklands-call-for-pilots-2025-deadline-extended/ [Accessed 17 Mar. 2026].